

A — device proposes (fast, NOT verified)
bf16 matmul scores $\rightarrow$ top- M candidate features

$\downarrow$ M ids

B — reference disposes (exact, verified)
recompute M scores, pinned math, policy-6 verdicts

$\downarrow$ facts

C — device applies (bit contract)
$$h' = \text{bf16}(\text{fp32}(\text{bf16}(h)) + \text{mag} \cdot \hat{n}_f)$$